1010 quiz1

1. An experiment was conducted to assess the effect of baking temperature on the density of bread. A large batch of white bread dough and a large batch of whole wheat bread dough were prepared. Each of the batches was formed into 18 loaves, resulting in a total of 36 loaves. Each of the white bread loaves was randomly assigned to one of three baking temperatures (low, medium, and high), so that 6 loaves were baked at the low temperature, 6 were baked at the medium temperature, and 6 were baked at the high temperature. A similar process of random assignment was carried out for the whole wheat bread loaves.

Which of the following are the treatments in the experiment?

(A) The 36 loaves of bread

(B) The type of bread (white, whole wheat)

(C) The density of the bread

(D) The baking temperatures (low, medium, high) ✓

Answer D

Correct. Baking temperature is an explanatory variable (or factor) in the experiment because the baking temperature is manipulated intentionally. The levels of the explanatory variable, the baking temperatures (low, medium, and high), are treatments.

2. For the purpose of determining the value of its end-of-year inventory, a clothing store creates a list at the end of the year of every item currently in stock along with each item's wholesale price. Which of the following is the best description of the end-of-the-year activity?

(A) An experiment, because the items are treatments and wholesale prices are responses.

(B) A sample survey, because the store wants to estimate the value of all items for the entire year.

(C) A sample survey, because the items currently in the store at the end of the year are a random sample of all items in the store for the entire year.

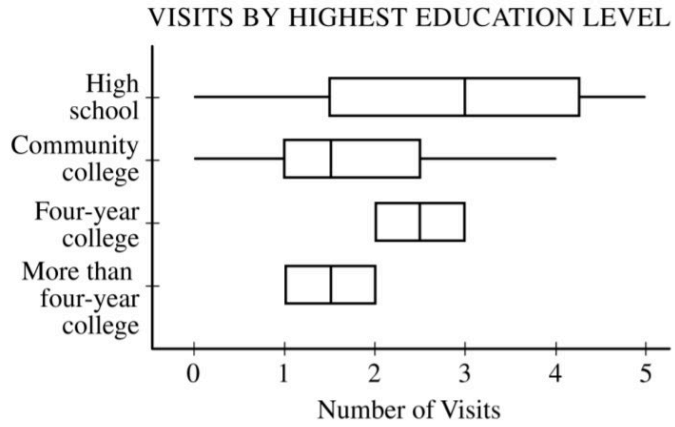
(D) A census, because the wholesale prices of all items are listed. ✓

Answer D

Correct. The end-of-year activity described is a census, since a list is made of every item in stock along with its corresponding wholesale price.

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3. A sociologist collected data from a sample of people on their highest level of education and the number of times they visited any fast food restaurant during the previous week. The data are summarized in the boxplots.



Based on the boxplots, which of the following statements must be true?

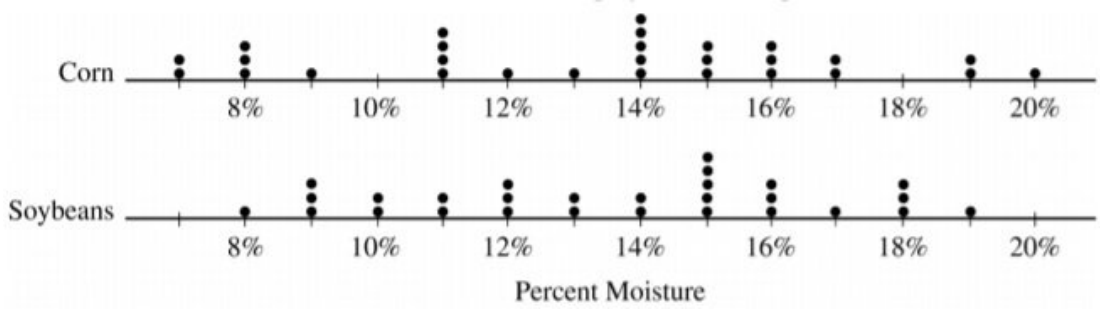
(A) The proportion of people surveyed from the first quartile to the third quartile at the four-year college level is less than the respective proportion at the community college level.

(B) The interquartile range (IQR) for the number of visits at the more than four-year college level is less than the IQR for the number of visits at the community college level. ✓

(C) The maximum number of visits at the community college level is greater than the maximum number of visits at the high school level.

(D) The median number of visits at the four-year college level is greater than the median number of visits at the high school level.

4. Grain moisture is a characteristic of grain that affects the price paid for the grain. A random sample of 28 loads of corn was evaluated for moisture as a percent of the total weight. A different random sample of 28 loads of soybeans was also evaluated for moisture. The data are displayed in the dotplots below.



Based on the dotplots, which of the following is greater for the percent moisture of corn than for the percent moisture of soybeans?

(A) The first quartile

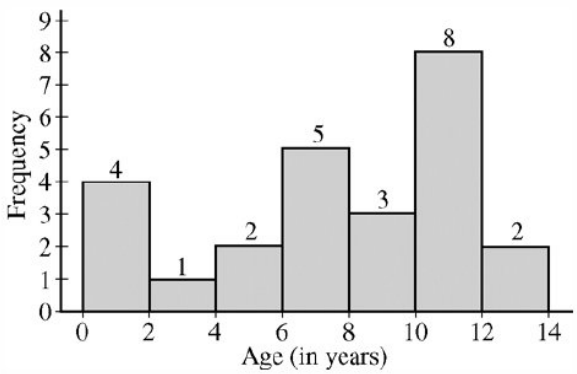
(B) The median

(C) The third quartile

(D) The range ✓

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5. A random sample of 25 households from the Mountainview School District was surveyed. In this survey, data were collected on the age of the youngest child living in each household. The histogram below displays the data collected in the survey.



In which of the following intervals is the median of these data located?

(A) 4 years old to less than 6 years old

(B) 6 years old to less than 8 years old

(C) 8 years old to less than 10 years old ✓

(D) 10 years old to less than 12 years old

6. For a recent season in college football, the total number of rushing yards for that season is recorded for each running back. The mean number of rushing yards for the running backs that season is 790 yards. One running back had 1,637 rushing yards for the season, which is 2.42 standard deviations above the mean number of rushing yards.

What is the standard deviation of the number of rushing yards for the running backs that season?

(A) 250 yards

(B) 300 yards

(C) 350 yards ✓

(D) 400 yards

7. A compact disc (CD) manufacturer wanted to determine which of two different cover designs for a newly released CD will generate more sales. The manufacturer chose 70 stores to sell the CD. Thirty-five of these stores were randomly assigned to sell CDs with one of the cover designs and the other 35 were assigned to sell the CDs with the other cover design. The manufacturer recorded the number of CDs sold at each of the stores and found a significant difference between the mean number of CDs sold for the two cover designs. Which of the following gives the conclusion that should be made based on the results and provides the best explanation for the conclusion?

(A) It is **not** reasonable to conclude that the difference in sales was caused by the different cover designs because this was not an experiment.

(B) It is **not** reasonable to conclude that the difference in sales was caused by the different cover designs because there was no control group for comparison.

(C) It is **not** reasonable to conclude that the difference in sales was caused by the different cover designs because the 70 stores were not randomly chosen.

(D) It is reasonable to conclude that the difference in sales was caused by the different cover designs because the cover designs were randomly assigned to stores. ✓

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8. For a sample of 42 rabbits, the mean weight is 5 pounds and the standard deviation of weights is 3 pounds. Which of the following is most likely true about the weights for the rabbits in this sample?

(A) The distribution of weights is approximately normal because the sample size is 42, and therefore the central limit theorem applies.

(B) The distribution of weights is approximately normal because the standard deviation is less than the mean.

(C) The distribution of weights is skewed to the right because the least possible weight is within 2 standard deviations of the mean. ✓

(D) The distribution of weights is skewed to the left because the least possible weight is within 2 standard deviations of the mean.

9. The number of hurricanes reaching the East Coast of the United States was recorded for each of the last ten decades by the National Hurricane Center. Summary measures are shown below.

Min = 12	Max = 24
Lower quartile = 15	Upper quartile = 18
Median = 16	n = 10

Which of the following statements is true?

(A) The largest observation is 24 and it is an outlier. No other observations in the data set could be outliers.

(B) Both 12 and 24 are outliers. It is possible that there are also other outliers.

(C) 12 is an outlier and it is possible that there are other outliers at the low end of the data set. There are no outliers at the high end of the data set.

(D) 24 is an outlier and it is possible that there are other outliers at the high end of the data set. There are no outliers at the low end of the data set. ✓

10. Administrators at a state university computed the mean GPA (grade point average) for juniors and seniors majoring in either physics or chemistry. The results are displayed in the table below. When juniors and seniors are grouped together, could physics majors have a higher mean GPA than chemistry majors?

	Physics	Chemistry
Juniors	2.8	3.0
Seniors	3.2	3.6
Overall	?	?

(A) No. The physics majors' mean GPA for juniors and seniors must be 3.0, while the chemistry majors' mean GPA for juniors and seniors must be 3.3.

(B) No. There is not enough information to determine the mean GPA for each major, but it must be higher for chemistry majors than for physics majors.

(C) Yes. It could happen. Whether it does happen depends on the number of juniors and seniors in each major. ✓

(D) Yes. It could happen. Whether it does happen depends on the variability of the GPAs within each of the four groups of students.

11. A data set of test scores is being transformed by applying the following rule to each of the raw scores.

$$\text{Transformed score} = 3.5(\text{raw score}) + 6.2$$

Which of the following is NOT true?

(A) The mean transformed score equals 3.5(the mean raw score) + 6.2.

(B) The range of the transformed scores equals 3.5(the range of the raw scores) + 6.2. ✓

(C) The standard deviation of the transformed scores equals 3.5(the standard deviation of the raw scores).

(D) The IQR of the transformed scores equals 3.5(the IQR of the raw scores).

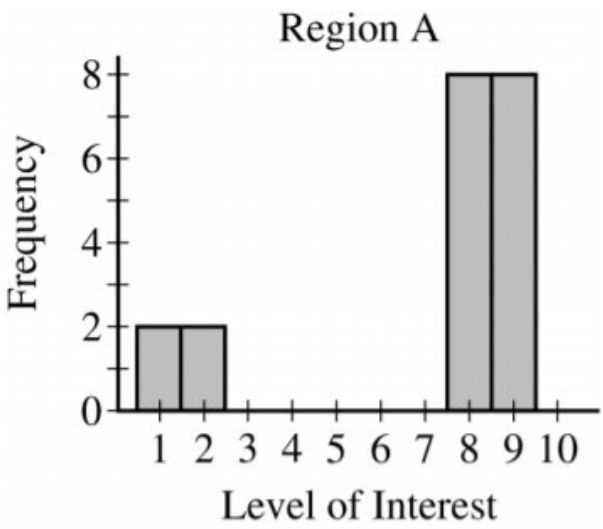
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12. A marketing firm obtained random samples of 20 people in five regions of the country to investigate the level of interest in a new product. People in the sample were asked to rate their level of interest on a scale from 1 to 10, with 1 being the least amount of interest and 10 being the greatest. The histograms show the results for each region.

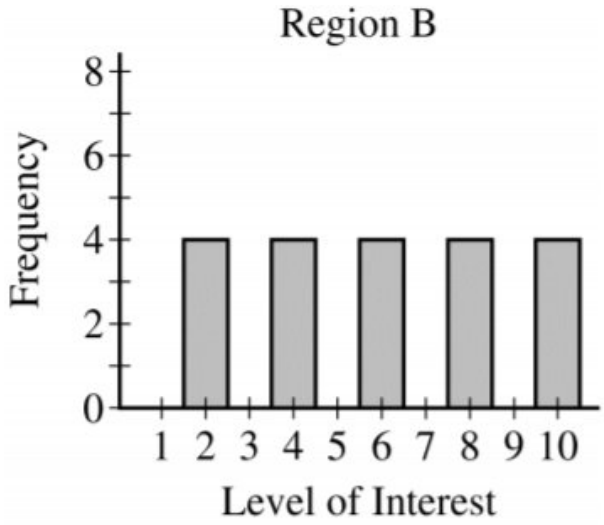
The graph for which region displays data for level of interest with the least standard deviation?

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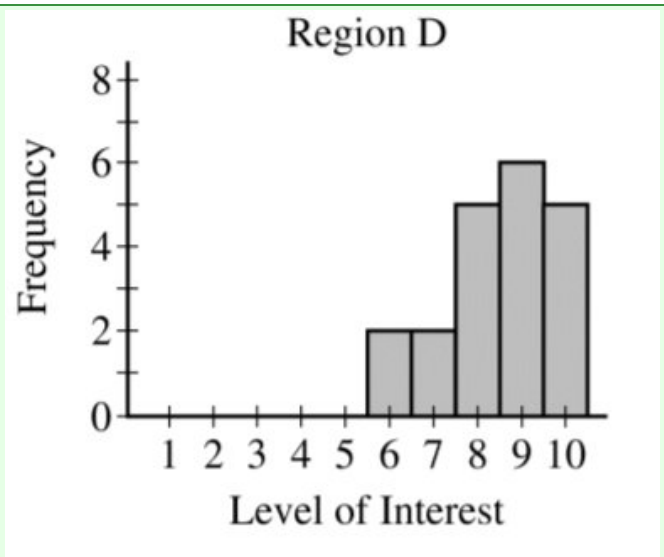
(A)



(B)

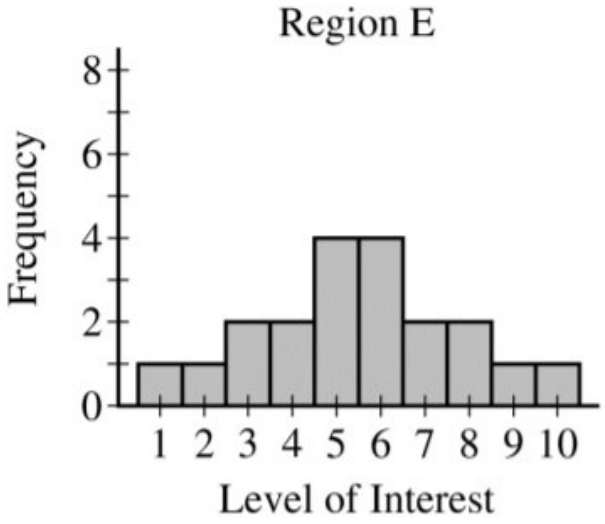


(C) ✓



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(D)



13. A new restaurant is interested in determining the best time-temperature combination for roasting a five-pound cut of lamb. The times to be tested are 45 minutes, 60 minutes, and 90 minutes at temperatures of 350 degrees Fahrenheit and 425 degrees Fahrenheit for each time, with the exception of the 90 minute - 425 degree combination. That combination is being eliminated because it will overcook the lamb, which leaves five combinations remaining. From 10 identical cuts of lamb, 2 are randomly selected to roast using each of the time-temperature combinations in the same oven. The quality of the finished product is evaluated for each roast. Which of the following is true?

(A) The explanatory variable is the quality of the finished lamb.

(B) If the experiment is repeated, identical results will be expected.

(C) There should be a control group (i.e., a group in which no treatment is given).

(D) The two cuts that are being roasted for each time-temperature combination are an example of replication. ✓

14. The final scores for 309 participants in a national golf tournament are summarized in the following table.

N	Mean	SE Mean	St Dev	Min	Q1	Median	Q3	Max
309	149.35	0.37	6.42	131	145	150	153.5	168

Which of the following is closest to the z-score of the minimum golf score?

(A) -2.16

(B) -2.86 ✓

(C) -2.96

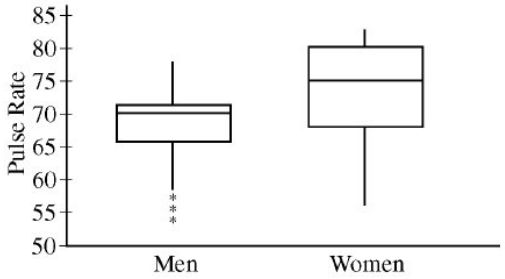
(D) -18.40

Answer B

Correct. The formula for a z-score is given by $z = \frac{x-\mu}{\sigma}$. For the minimum golf score, the z-score is equal to $\frac{131-149.35}{6.42}$, which is approximately equal to -2.86.

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15. The pulse rate for each person in a sample of 20 men and 20 women was recorded. The boxplots below summarize the pulse rates for the men and the women in the sample.



Which of the following statements about the people in the sample must be true?

(A) There are more people between the first and third quartiles for women than there are between the first and third quartiles for men.

(B) At least half of the women had higher pulse rates than three-fourths of the men. ✓

(C) More than half of the men had lower pulse rates than three-fourths of the women.

(D) If a man and a woman were randomly selected from the 40 people, the man would have the lower pulse rate.

16. An athletic trainer believes that a new natural treatment will decrease the mean healing time, in days, compared to the current treatment for athletes with knee injuries. The athletic trainer would like to create a study to test his belief. The mean healing time for athletes receiving the current treatment is 21 days. An investigative question will be formulated before the study is created.

Which of the following would be the appropriate components of the athletic trainer's investigative question?

(A) The parameter is the mean healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is not equal to 21.

(B) The parameter is the mean healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is less than 21. ✓

(C) The parameter is the healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is not equal to 21.

(D) The parameter is the healing time for knee injuries when using the new natural treatment, and the direction of the alternative hypothesis is less than 21.

Answer B

Correct. Mean healing time is the parameter. The direction of the alternative hypothesis is correct because the direction is less than 21.

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17. Heights, in inches, for the 200 graduating seniors from Washington High School are summarized in the frequency table below.

Height	Frequency
Height < 60 inches	22
60 inches ≤ height < 66 inches	84
66 inches ≤ height < 72 inches	62
72 inches ≤ height < 78 inches	24
Height ≥ 78 inches	8

Which of the following statements about the median height is true?

(A) It is greater than or equal to 72 inches but less than 78 inches.

(B) It is greater than or equal to 66 inches but less than 72 inches.

(C) It is greater than or equal to 60 inches but less than 66 inches. ✓

(D) It is less than 60 inches.

18. A doctor wanted to publish a research study about newborn babies in a certain city. She randomly selected newborn babies from the city and recorded their weights, in kilograms, using the same scale. She later discovered that the scale she used was incorrectly adding 0.25 kilogram to the weight of each newborn baby. To correct the error, she adjusted each of the weights in the random sample by subtracting 0.25 kilogram.

Which of the following statistics for the weights, in kilograms, of the newborn babies has the same value before and after the doctor's adjustment?

(A) Median

(B) Mean

(C) Third quartile

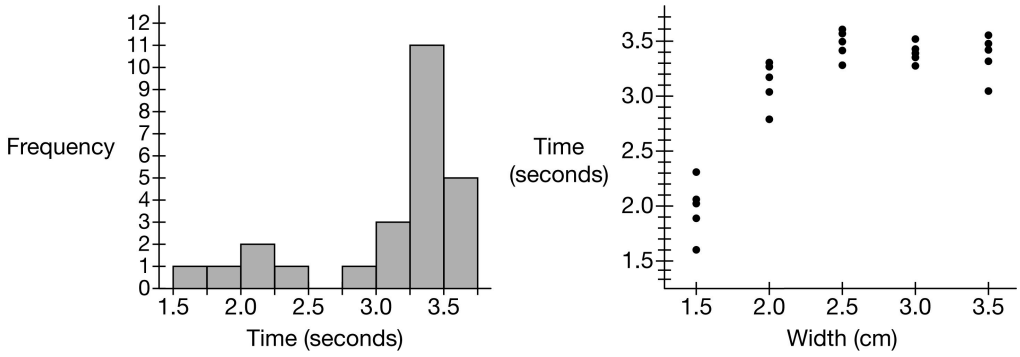
(D) Interquartile range ✓

Answer D

Correct. The interquartile range would not change if the weights were adjusted because it does not change the difference between the third and first quartiles.

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19. Students in a physics class designed an experiment to investigate the effect of blade width on flight time for paper helicopters. A histogram of the flight times, in seconds, and a scatterplot of flight time versus blade width, in centimeters (cm), are shown for 25 helicopters.



Which of the following best describes the clusters and the gap in the histogram?

(A) The left cluster displays flight times for blade widths from 1.5 cm to 2.5 cm, and the right cluster displays flight times for the remaining blade widths.

(B) The left cluster displays flight times for blade widths of 1.5 cm, and the right cluster displays flight times for the remaining blade widths. ✓

(C) The clusters are not related to the blade widths because flight time is measured in seconds and blade widths are measured in centimeters.

(D) The clusters are not related to the blade widths because the relationship between blade width and flight time is not linear.

Answer B

Correct. The histogram indicates that there are no observed data values between 2.5 and 2.75 seconds, with one cluster less than 2.5 seconds and another cluster greater than 2.75 seconds. The scatterplot indicates that there are 5 observations less than 2.5 seconds (the left cluster) and each of these observations has a blade width of 1.5 cm.

20. A study was conducted to evaluate the impact of taking a nutritional supplement on a person's reaction time. One hundred volunteers were placed into one of three groups according to their athletic ability: low, moderate, or high. Participants in each group were randomly assigned to take either the nutritional supplement or a placebo for six weeks. At the end of the six weeks, participants were given a coordination task. The reaction time in completing the task was recorded for each participant. The study compared the reaction times between those taking the supplement and those taking the placebo within each athletic ability level.

Which of the following is the best description of the study?

(A) A randomized block design ✓

(B) A completely randomized design

(C) A matched-pairs design

(D) A randomized observational study

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21. The transportation department of a large city wants to estimate the proportion of residents who would use a system of aerial gondolas to commute to work. The gondolas would be part of the city's effort to relieve traffic congestion. The department asked a random sample of residents whether they would use the gondolas. The residents could respond with yes, no, or maybe.

Which of the following is the best description of the method for data collection used by the department?

(A) A census

(B) A sample survey ✓

(C) An experiment with a completely randomized design

(D) An experiment with a randomized block design

22. A researcher wanted to estimate the average amount of money spent on extracurricular activities per school in a certain region. The researcher randomly selected 20 public schools and 20 private schools in the region to use for a sample. Which of the following best describes the type of sample that was taken?

(A) A cluster sample

(B) A convenience sample

(C) A simple random sample

(D) A stratified random sample ✓